

PugaAI Health

NAIC Validation & Admin Console

Product Requirements Document — PRD v1.0 Addendum

Product: PugaAI Health

Parent PRD: PugaAI Health Product Requirements Document v1.0

Module: NAIC Validation & Admin Console

Release Target: PugaAI Health v1.4.2+

Document Status: Developer-Ready

Primary Stack: React + Vite frontend, Express.js backend

Database: PostgreSQL or approved production relational database

NAIC Problem Statement: 02 — Voice-First Access

Purpose: Capture, protect, review and export evidence of genuine PugaAI Health voice interactions for NAIC validation.

1. Purpose

This PRD defines the administrative validation layer required to support PugaAI Health's participation in the National AI Innovation Challenge (NAIC).

The module extends the existing PugaAI Health product without replacing the existing patient-facing experience.

The module provides:

1. Secure administrator authentication.
2. Role-based access control.
3. Automatic logging of completed voice interactions.
4. Pseudonymized participant identification.
5. NAIC validation metrics.
6. Interaction review.
7. Validation progress toward the 50-interaction requirement.
8. CSV evidence export.
9. Audit logging.
10. Privacy and data-retention controls.

The NAIC requirements state that Voice-First Access submissions require at least 50 documented real user interactions and genuine integration with the official N-ATLAS ASR service. Therefore, this module must produce evidence from actual system interactions rather than manually entered or simulated records.

2. Relationship to Existing PugaAI Health PRD

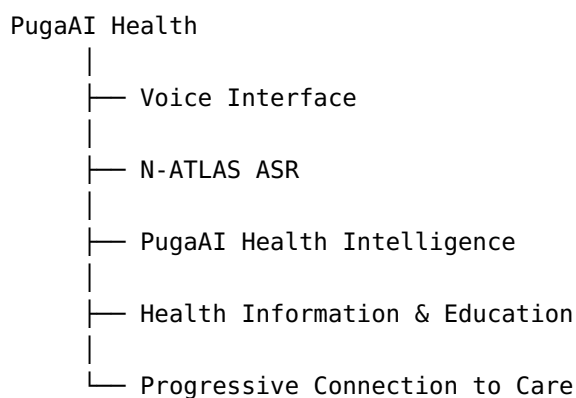
This document is an **addendum** to the existing PugaAI Health PRD v1.0.

It does not replace or modify the core product architecture unless explicitly stated.

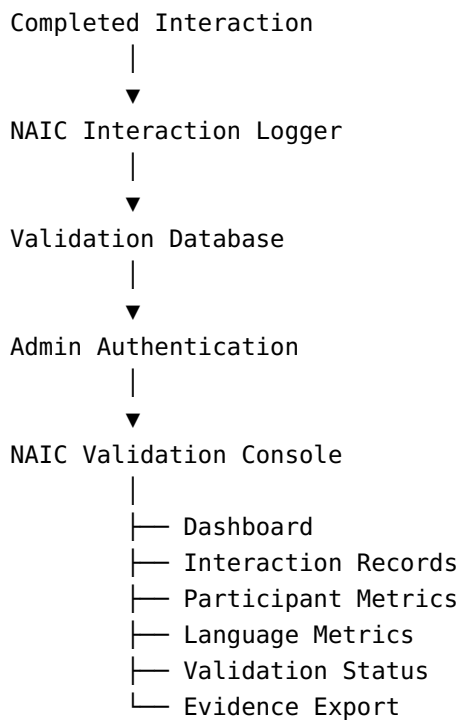
The existing PugaAI Health product remains:

PugaAI Health — Your AI-powered voice health navigator.

The existing patient-facing architecture remains unchanged:



The new administrative architecture adds:



3. Problem Statement

PugaAI Health needs to demonstrate real-world use of its N-ATLAS-powered voice-first health-information service.

The product must therefore capture sufficient evidence to demonstrate:

- actual user interaction;
- actual voice input;
- N-ATLAS processing;
- successful PugaAI Health response;
- supported language/channel;
- validation outcome;
- user feedback where collected;
- date/time;
- pseudonymized participant reference.

The existing Puga TriniCare Super Admin Dashboard is not a prerequisite for this validation.

A lightweight, purpose-built NAIC Validation Console is therefore required.

4. Goals

4.1 Primary Goals

The system shall:

1. Automatically record eligible completed voice interactions.
2. Associate each interaction with a pseudonymized participant.
3. Record the relevant N-ATLAS processing evidence.
4. Provide an administrator-only dashboard.
5. Track progress toward the minimum 50 documented interactions.
6. Provide interaction-level review.
7. Provide CSV export for NAIC evidence preparation.
8. Protect validation records from ordinary users.
9. Maintain an audit trail of administrator access.
10. Support later migration into the full Puga TriniCare Super Admin Dashboard.

4.2 Secondary Goals

The system should:

- support multiple languages;
- support web voice initially;
- support PugaAccess channels later;
- support filtering;
- support validation quality metrics;

- support future evidence packages;
 - minimize unnecessary personal data collection.
-

5. Non-Goals

This module shall NOT:

- replace the Puga TriniCare Super Admin Dashboard;
 - provide patient medical-record administration;
 - diagnose patients;
 - prescribe medication;
 - make clinical decisions;
 - expose protected patient health records to NAIC administrators by default;
 - become a general analytics dashboard;
 - store raw audio by default;
 - allow administrators to edit historical interaction evidence;
 - allow fabricated/manual interactions to count toward the official validation total.
-

6. User Roles

6.1 Patient / General User

Permissions:

- Use PugaAI Health.
- Submit voice questions.
- Receive health information.
- Provide optional feedback.

Cannot:

- access NAIC Validation Console;
 - view other participants;
 - view validation statistics;
 - export interaction data.
-

6.2 Health Educator / Validation Coordinator

Purpose:

Support participant recruitment and supervised validation activities.

Default permissions:

- recruit participants;

- create/start validation sessions where authorized;
- assist participants with onboarding;
- view only information necessary for their assigned validation activity.

The role should NOT automatically receive access to all historical validation records.

Access should be configurable.

6.3 NAIC Administrator

Permissions:

- Login to NAIC Administration.
- View validation dashboard.
- View interaction records.
- Filter records.
- View permitted interaction evidence.
- View validation metrics.
- Export validation dataset.
- View validation audit events.

Cannot:

- alter completed evidence records;
 - alter N-ATLAS transcripts;
 - alter user responses;
 - delete historical evidence without elevated authorization.
-

6.4 Super Administrator

Permissions:

- All NAIC Administrator permissions.
 - Manage NAIC administrator accounts where implemented.
 - Configure validation settings.
 - Configure retention policy.
 - Review audit logs.
 - Disable validation access.
 - Perform authorized administrative actions.
-

7. Authentication Requirements

The NAIC Console must require authentication.

The console must never be publicly accessible.

Required flow:

```
/admin/naic
  ↓
Authentication Check
  ↓
Not authenticated → Login
  ↓
Authenticated
  ↓
Role Check
  ↓
naic_admin / super_admin
  ↓
NAIC Console
```

Unauthorized roles must receive:

```
HTTP 403 Forbidden
```

The backend, not the frontend, is the authoritative authorization boundary.

8. Admin Login

8.1 Required Fields

- Email or username
- Password

8.2 Recommended Future Enhancement

Multi-factor authentication.

MFA is recommended for production administrative access but is not required to block the initial NAIC validation build.

8.3 Session Requirements

Admin sessions must:

- expire;
- support logout;
- not share patient authentication state;
- use secure authentication mechanisms;
- avoid storing secrets in insecure browser-accessible locations;

- invalidate the session on logout.
-

9. NAIC Validation Session

A validation session identifies the participant without unnecessarily collecting personal information.

Required participant identifier

Example:

```
P-001  
P-002  
P-003
```

The participant identifier must be pseudonymous.

Do not use:

- full name;
- phone number;
- email;
- National Identification Number;
- health ID;

unless there is a separately justified and consented requirement.

10. Consent

Before a validation interaction is counted, the system should establish that the participant has received the appropriate validation/testing notice and provided the required consent.

Minimum conceptual states:

```
CONSENT_PENDING  
CONSENT_GRANTED  
CONSENT_DECLINED
```

An interaction associated with `CONSENT_DECLINED` must not be included in the official validation count.

Consent implementation must comply with the applicable Puga privacy/data-protection framework.

11. Interaction Definition

A valid NAIC interaction is a genuine user voice interaction that reaches the intended PugaAI Health processing flow.

Minimum successful chain:

```
Real Participant
  ↓
Voice Input
  ↓
N-ATLAS ASR
  ↓
Transcript
  ↓
PugaAI Health Processing
  ↓
PugaAI Health Response
```

A microphone button click alone does NOT count.

A failed request does NOT automatically count.

A simulated test record does NOT count.

An administrator-created record does NOT count.

12. Interaction Record

Each eligible interaction should contain the following structured fields.

Identity

```
interaction_id
participant_id
validation_session_id
```

Timing

```
started_at
completed_at
duration_ms
```

Channel

```
channel
```

Examples:

```
web_voice  
mobile_voice  
pugaaccess_ivr  
pugaaccess_usd  
pugaaccess_sms
```

Language

```
language
```

Initial supported values:

```
en-NG  
yo-NG  
ig-NG  
ha-NG
```

N-ATLAS Evidence

```
n_atlas_used  
n_atlas_service  
transcript  
asr_confidence
```

Where available:

```
n_atlas_request_id
```

PugaAI Processing

```
conversation_id  
message_id  
response_status  
response_category
```

Possible response category:

```
health_information
health_education
prevention
warning_signs
care_navigation
safety_escalation
```

Validation

```
validation_status
user_feedback
feedback_rating
```

Possible status:

```
completed
failed
partial
retry
excluded
```

13. Raw Audio

Raw voice recordings must NOT be retained by default.

The default evidence model should retain:

- interaction metadata;
- transcript;
- N-ATLAS processing metadata;
- PugaAI response metadata;
- validation outcome.

If audio retention is introduced later, it requires:

1. explicit product decision;
2. appropriate participant consent;
3. defined retention period;
4. encrypted storage;
5. strict access control;
6. audit logging;
7. deletion process.

14. Automatic Logging

When a qualifying interaction successfully completes, the frontend/API integration shall submit the evidence record.

Preferred flow:

```
Voice interaction completes
    ↓
PugaAI response succeeds
    ↓
Validation logger triggered
    ↓
POST /api/v1/naic/interactions
    ↓
Express validation service
    ↓
Database
```

The logging operation must not expose internal database credentials to the browser.

15. Backend API

15.1 Create Interaction

```
POST /api/v1/naic/interactions
```

Example payload:

```
{
  "participantId": "P-001",
  "validationSessionId": "VS-0001",
  "channel": "web_voice",
  "language": "en-NG",
  "nAtlasUsed": true,
  "transcript": "What are the warning signs of malaria?",
  "asrConfidence": 0.94,
  "conversationId": "conv_123",
  "messageId": "msg_456",
  "responseStatus": "completed",
  "responseCategory": "health_information",
}
```

```
"validationStatus": "completed"
}
```

The server must generate:

```
interaction_id
created_at
```

The client must not be trusted to determine whether an interaction is genuinely eligible.

16. Validation Eligibility Rules

An interaction counts toward the official validation count only when:

```
participant exists
AND
consent = granted
AND
voice input was submitted
AND
N-ATLAS ASR was used
AND
transcript exists
AND
PugaAI response completed
AND
validation status = completed
```

The backend should determine eligibility.

Recommended field:

```
counts_toward_validation: true | false
```

This field must be server-generated.

17. Validation Dashboard

Route:

```
/admin/naic
```

Dashboard sections:

17.1 Progress

NAIC Voice Validation

57 / 50 interactions
114% of target

The target should be configurable but default to:

50

17.2 Key Metrics

Cards:

- Total valid interactions
- Unique participants
- Languages tested
- Successful interactions
- Failed interactions
- Completion rate

18. Language Breakdown

Example:

Nigerian English	42
Yorùbá	8
Igbo	5
Hausa	5

The dashboard must derive this from stored records.

19. Channel Breakdown

Example:

Web Voice	45
IVR	8
USSD	4
SMS	3

Only channels actually integrated and used should appear as completed evidence.

20. Interaction Table

Columns:

Interaction ID
Participant ID
Language
Channel
N-ATLAS
Status
Category
Timestamp

Example:

Interaction	Participant	Language	Channel	N-ATLAS	Status
NAI-0001	P-001	en-NG	Web Voice	Yes	Complete
NAI-0002	P-001	en-NG	Web Voice	Yes	Complete
NAI-0003	P-002	yo-NG	Web Voice	Yes	Complete

21. Interaction Detail

Selecting an interaction opens a protected detail view.

Display:

Metadata

- Interaction ID
- Participant ID
- Date/time
- Language
- Channel
- Validation status

N-ATLAS

- N-ATLAS used
- transcript
- ASR confidence
- request reference where permitted

PugaAI Health

- conversation reference
- response category
- response status
- safety classification where applicable

Feedback

- user rating
- user feedback
- validation notes where authorized

The administrator must not be able to modify the original system-generated evidence.

22. Filtering

Required filters:

- Date range
- Language
- Channel
- Validation status
- Response category
- Participant ID

Optional:

- N-ATLAS success
 - feedback rating
-

23. CSV Export

Endpoint:

```
GET /api/v1/naic/interactions/export
```

Export file:

PugaAI_NAIC_Validation_Interactions.csv

Recommended fields:

```
interaction_id
participant_id
validation_session_id
started_at
completed_at
channel
language
n_atlas_used
transcript
asr_confidence
response_category
response_status
validation_status
feedback_rating
counts_toward_validation
```

The export must exclude unnecessary sensitive personal data.

24. Export Authorization

Only:

```
naic_admin
super_admin
```

may export NAIC validation evidence.

All exports should generate an audit event:

NAIC_EXPORT_CREATED

with:

```
admin_id
timestamp
filters
record_count
```

25. Audit Logging

Administrative actions must be auditable.

Required events include:

```
ADMIN_LOGIN_SUCCESS
ADMIN_LOGIN_FAILURE
ADMIN_LOGOUT
NAIC_DASHBOARD_VIEW
NAIC_INTERACTION_VIEW
NAIC_EXPORT_CREATED
NAIC_VALIDATION_SETTINGS_CHANGED
NAIC_ADMIN_ACCESS_DENIED
```

Audit records should include:

```
event_id
admin_id
event_type
timestamp
request_id
result
```

Do not store passwords or authentication secrets in audit logs.

26. Security Requirements

The backend must enforce:

```
naic_admin → allowed
super_admin → allowed
patient → forbidden
health_educator → restricted
anonymous → forbidden
```

Requirements:

- server-side authorization;
- secure session/token handling;
- HTTPS in production;
- rate limiting on authentication;
- secure password storage;
- no plaintext passwords;

- no API credentials in frontend source;
 - no sensitive information in client-side error messages;
 - audit logging;
 - least-privilege access;
 - secure CSV generation;
 - protection against IDOR;
 - protection against unauthorized interaction lookup.
-

27. Data Protection

The NAIC module must follow the principle of data minimization.

The validation dataset should answer:

“Can PugaAI Health demonstrate genuine voice-first usage with N-ATLAS?”

without unnecessarily answering:

“Who exactly is this person and what is their medical history?”

Therefore:

Validation evidence ≠ Patient medical record

The Puga Universal Health ID must not be required merely to satisfy NAIC interaction validation.

28. Relationship to Puga Universal Health ID

The NAIC validation system must remain independent from protected patient identity.

A validation participant does not need a Puga Universal Health ID solely for testing.

If a participant is an existing Puga user, the validation system should preferably use a pseudonymous validation participant identifier rather than exposing the user's Health ID in the NAIC dataset.

29. Health Safety

The NAIC validation system must not change the clinical safety rules of PugaAI Health.

Phase 1 remains focused on:

- primary health information;

- health education;
- prevention;
- warning signs;
- care navigation;
- appropriate escalation.

PugaAI Health must not independently:

- diagnose;
- prescribe;
- make irreversible clinical decisions.

30. Progressive Product Activation

The NAIC module must work even when PugaCare/PugaCure/PugaPay/PugaAccess are not fully integrated.

Initial capability configuration may be:

```
AI_HEALTH_INFORMATION = ON
VOICE_INTERFACE = ON
N_ATLAS_ASR = ON
NAIC_VALIDATION = ON

CARE_FINDER = OFF
APPOINTMENTS = OFF
TELECONSULTATION = OFF
LABORATORY = OFF
PRESCRIPTIONS = OFF
PUGAPAY = OFF
PUGAACCESS = OFF
```

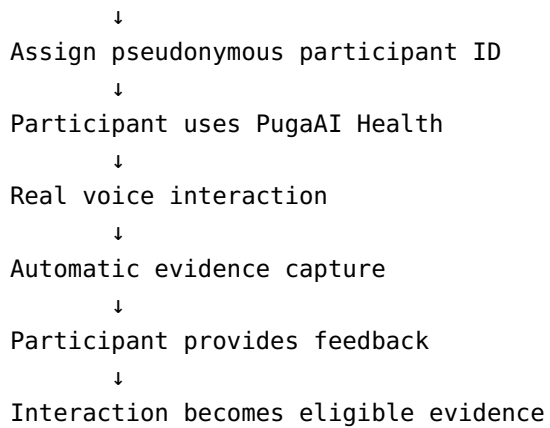
Capabilities can subsequently be activated individually.

31. Health Educator Validation Workflow

The Health Educator Exchange Program associate may act as a validation coordinator.

Workflow:

```
Recruit participant
  ↓
Explain validation
  ↓
Obtain appropriate consent
```



The coordinator must not manufacture, replay or fabricate interactions.

32. Validation Target

Official NAIC target:

Minimum = 50 documented real user interactions

Recommended operational target:

60+ interactions

The additional buffer protects against:

- invalid interactions;
- failed interactions;
- incomplete records;
- excluded records;
- technical failures.

The system should display:

Valid interactions / target

rather than simply counting all microphone attempts.

33. Example Validation Plan

Recommended initial recruitment:

```
20 participants
×
3 genuine voice interactions
=
60 interactions
```

Participants may use realistic primary-health-information questions.

Examples:

- malaria prevention;
- child immunization information;
- antenatal-care information;
- hypertension prevention;
- warning signs requiring urgent care;
- general preventive health.

Questions must represent genuine user interactions rather than artificially generated evidence.

34. Frontend Routes

Required routes:

```
/admin/naic/login
/admin/naic
/admin/naic/interactions/:interactionId
```

Optional future routes:

```
/admin/naic/participants
/admin/naic/audit
/admin/naic/settings
```

35. Frontend Components

Required components:

```
NAICAdminLogin
NAICValidationDashboard
NAICProgressCard
NAICMetricCard
NAICLanguageBreakdown
```

```
NAICChannelBreakdown
NAICInteractionTable
NAICInteractionDetail
NAICExportButton
NAICAdminHeader
NAICProtectedRoute
```

36. Backend Services

Recommended Express services:

```
AdminAuthService
NAICValidationService
NAICInteractionService
NAICExportService
NAICAuditService
```

Recommended middleware:

```
requireAuthentication
requireRole
requireNAICAdmin
validateNAICInteraction
```

37. Suggested Database Tables

`admin_users`

```
id
email
password_hash
role
status
created_at
updated_at
last_login_at
```

`naic_validation_sessions`

```
id
participant_id
```

```
consent_status
coordinator_id
started_at
ended_at
status
```

naic_interactions

```
id
validation_session_id
participant_id
channel
language
n_atlas_used
n_atlas_request_id
transcript
asr_confidence
conversation_id
message_id
response_category
response_status
validation_status
feedback_rating
counts_toward_validation
started_at
completed_at
created_at
```

admin_audit_events

```
id
admin_id
event_type
request_id
metadata
created_at
```

38. API Error Standards

Examples:

```
401 UNAUTHORIZED
403 FORBIDDEN
404 NOT_FOUND
```

```
409 DUPLICATE_INTERACTION
422 VALIDATION_ERROR
429 RATE_LIMITED
500 INTERNAL_ERROR
```

User-facing responses must not expose database errors, stack traces or internal infrastructure details.

39. Duplicate Protection

The backend should protect against accidental duplicate submissions.

Recommended:

```
idempotency_key
```

or an equivalent server-side deduplication mechanism.

The same completed voice interaction must not be counted twice because of:

- browser retry;
- network retry;
- page refresh;
- duplicate API submission.

40. Reliability Requirements

If the validation logging request temporarily fails:

- the user should still receive the PugaAI response where safe;
- the frontend should safely retry according to the existing API retry policy;
- duplicate records must be prevented;
- the system should surface logging failure to the validation environment;
- the interaction should not falsely appear as successfully documented if the server never persisted it.

41. Analytics

The console should calculate:

```
total_valid_interactions
unique_participants
completion_rate
```

```
failure_rate
language_distribution
channel_distribution
daily_interactions
feedback_distribution
```

Optional:

```
average_asr_confidence
average_interaction_duration
```

Analytics must be derived from actual records.

42. NAIC Evidence Package

The module should support preparation of the following evidence:

Evidence 1 — Working Application

PugaAI Health live voice interaction.

Evidence 2 — N-ATLAS Integration

Technical documentation demonstrating N-ATLAS ASR integration.

Evidence 3 — Real-World Validation

Exported interaction dataset demonstrating 50+ genuine interactions.

Evidence 4 — Technical Documentation

Architecture/API/security documentation.

Evidence 5 — Demonstration Video

Show:

```
User speaks
→ N-ATLAS ASR
→ PugaAI Health
→ response
→ interaction logged
→ admin login
→ validation dashboard
```

- interaction evidence
- CSV export

The current NAIC submission requirements identify these categories as part of the application package.

43. Definition of Done

The module is complete when:

Authentication

- ☐ Admin login works.
- ☐ Admin logout works.
- ☐ Unauthorized users cannot access the console.
- ☐ Backend RBAC is enforced.
- ☐ Admin sessions expire appropriately.

Interaction Logging

- ☐ Genuine voice interactions are automatically logged.
- ☐ N-ATLAS usage is recorded.
- ☐ Transcript is captured.
- ☐ Participant is pseudonymized.
- ☐ Duplicate submissions are prevented.
- ☐ Invalid interactions do not count.

Dashboard

- ☐ 50-interaction progress is displayed.
- ☐ Participant count is displayed.
- ☐ Language breakdown works.
- ☐ Channel breakdown works.
- ☐ Interaction table works.
- ☐ Interaction details work.

Evidence

- ☐ CSV export works.
- ☐ Export is restricted to authorized administrators.
- ☐ Export excludes unnecessary sensitive information.
- ☐ Export event is audited.

Security

- ☐ Server-side RBAC is implemented.
- ☐ IDOR protection is implemented.
- ☐ Sensitive errors are redacted.
- ☐ Audit logging works.
- ☐ Authentication rate limiting is implemented.

Validation

- [] Real participant workflow works.
 - [] Consent state is recorded.
 - [] 50+ genuine interactions can be documented.
 - [] Evidence can be independently reviewed.
-

44. Testing Requirements

Development must follow:

RED → GREEN → REFACTOR

Required tests include:

Unit Tests

- interaction eligibility;
- participant pseudonymization;
- validation counter;
- language aggregation;
- channel aggregation;
- duplicate protection;
- export transformation.

Integration Tests

- admin login;
- role authorization;
- interaction creation;
- validation retrieval;
- interaction detail;
- CSV export;
- audit event creation.

Security Tests

- patient attempting admin route;
- anonymous API request;
- invalid role;
- IDOR attempt;
- expired session;
- unauthorized export.

E2E Tests

Critical flow:

```
Admin login
→ NAIC dashboard
→ view interaction
→ filter interactions
→ export CSV
→ logout
```

45. Performance Requirements

Dashboard initial load target:

```
< 2 seconds
```

for normal validation datasets.

Interaction listing must support pagination.

Do not load thousands of interaction records into the browser simultaneously.

Recommended:

```
page
page_size
sort
filter
```

server-side query parameters.

46. Future Integration with Super Admin Dashboard

The NAIC Validation Console should be implemented as a modular service.

Later:

```
Puga TriniCare Super Admin Dashboard
|
├── Users
├── Patients
├── Facilities
└── Providers
```

```
├─ PugaAccess
├─ PugaPay
├─ PugaAI
└─ NAIC Validation
```

The NAIC module should therefore avoid architecture that prevents future migration.

47. Recommended Environment Variables

Frontend:

```
VITE_API_BASE_URL
VITE_NAIC_VALIDATION_ENABLED
```

Backend:

```
NAIC_VALIDATION_ENABLED
NAIC_VALIDATION_TARGET
NAIC_EXPORT_ENABLED
NAIC_SESSION_TIMEOUT
NAIC_AUDIO_RETENTION_ENABLED
```

Audio retention should default to:

```
false
```

48. Configuration

Initial configuration:

```
NAIC_VALIDATION_ENABLED=true
NAIC_VALIDATION_TARGET=50
NAIC_AUDIO_RETENTION_ENABLED=false
```

The target should not be hardcoded into the UI.

49. Release Strategy

Release 1 — NAIC Admin Foundation

- Admin login
- RBAC
- validation database
- interaction logger
- dashboard
- export

Release 2 — N-ATLAS Live Validation

- production N-ATLAS ASR
- real voice interaction logging
- validation sessions
- participant recruitment workflow

Release 3 — Evidence Hardening

- audit improvements
- validation reports
- stronger export package
- evidence integrity checks

Release 4 — Super Admin Integration

Move or surface NAIC functionality inside the broader Puga TriniCare Super Admin Dashboard.

50. Developer Handoff

The development team should treat this document as the authoritative specification for the **NAIC Validation & Admin Console**.

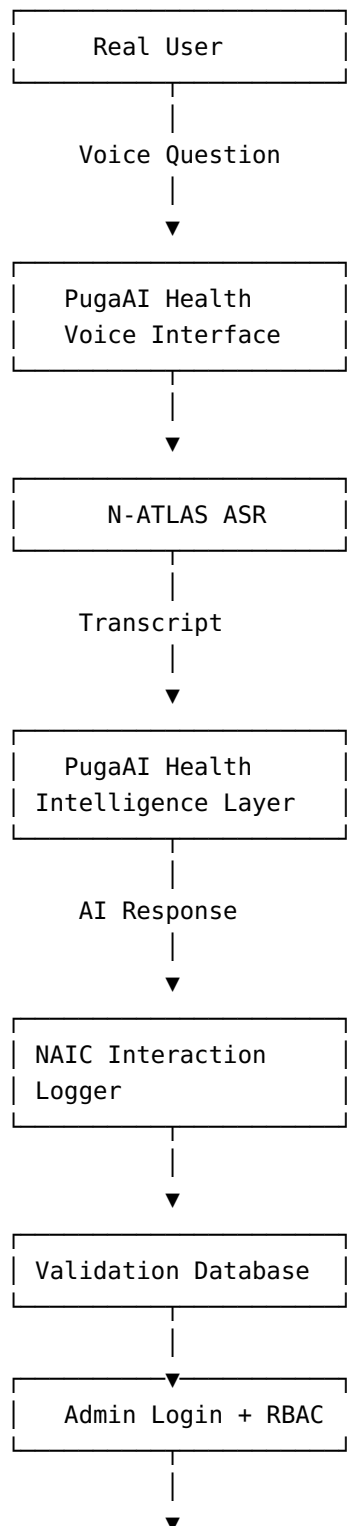
The existing PugaAI Health PRD remains authoritative for the main product.

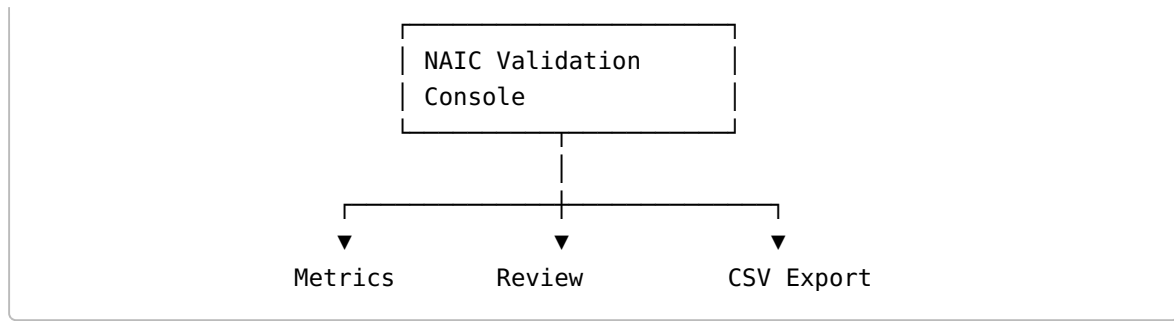
The implementation must preserve:

```
One PugaAI Health service
    +
Multiple access channels
    +
One shared backend
    +
Progressive capability activation
```

The NAIC Admin Console is an **administrative evidence layer**, not a second PugaAI Health implementation.

51. Core Architecture Summary





52. Final Product Principle

The system must make the following statement technically demonstrable:

A real Nigerian user spoke to PugaAI Health, the voice input was processed through N-ATLAS, PugaAI Health returned a response, and the interaction was securely documented for real-world validation.

That evidence must be generated by the system itself rather than manually assembled after the fact.